

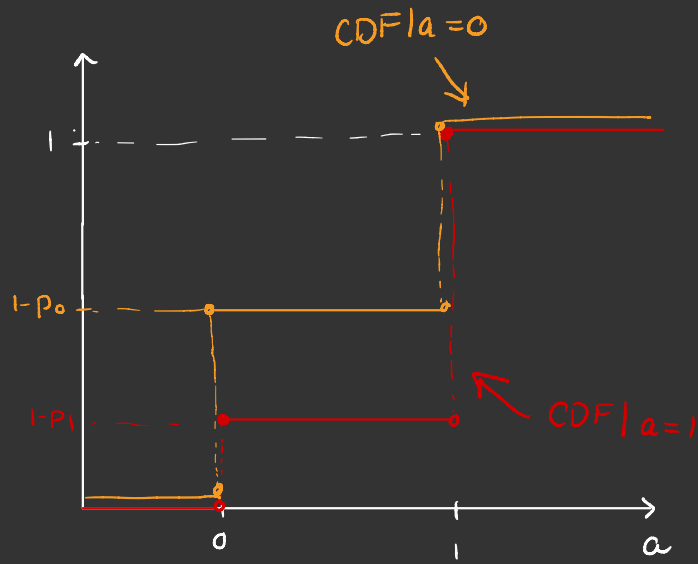
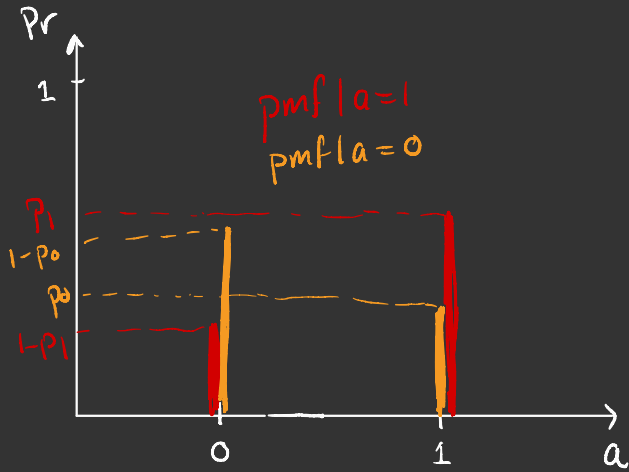
Lecture 8

Moral Hazard

ECON 403/503
Fall 2020



Comments on 10.



$$\therefore \text{Dist. under } a=1 \geq \text{Dist. under } a=0$$

FOSD

$\therefore$ effort generates a distribution that FOSD the one with no effort

Notes on 13:

From FOC on w_1 ,

$$\frac{p_1}{p_1 u'(w_1)} = \lambda \Leftrightarrow \frac{1}{u'(w_1)} = \lambda$$

$u'(\cdot) > 0$ for any w
so $u'(w_1) > 0$

$\Rightarrow \lambda > 0$
 $\Rightarrow$ From complementary slackness:
 $p_1 u(w_1) + (1-p_1) u(w_0) = \psi$ DC
is binding.

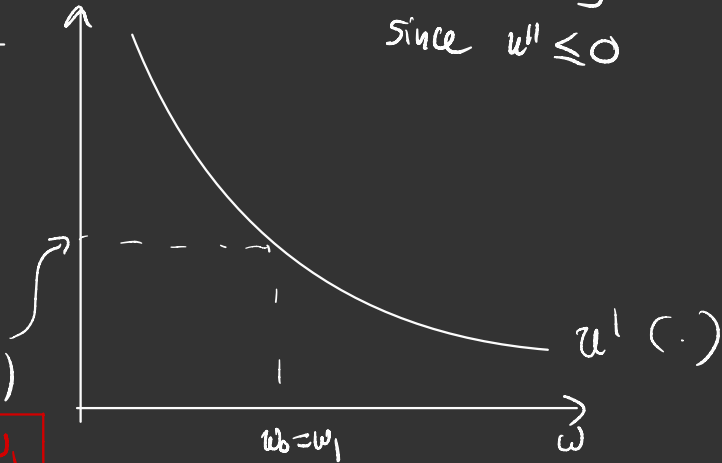
From FOC's on w_1 and w_2 ,
eliminating λ :

$$\frac{1}{u'(w_1)} = \lambda = \frac{1-p_1}{(1-p_1) u'(w_0)} \Rightarrow$$

$$\frac{1}{u'(w_1)} = \frac{1}{u'(w_0)} = \lambda \Rightarrow u'(w_0) = u'(w_1)$$

$$\Rightarrow \boxed{w_0 = w_1}$$

u' is decreasing monotonically
since $u'' \leq 0$



Notes on 15:

Since $(w_0^f, w_1^f) = (w^*, w^*)$, from IR constraint we get

that $u(w^*) = \psi$,

Let h denote the inverse of u : $h(x) = u^{-1}(x)$

$$\Rightarrow w^* = h(\psi) = u^{-1}(\psi)$$

w^* = cost in money of putting effort for the agent.

u : Wages $\rightarrow$ utility

u^{-1} : utils $\rightarrow$ wages

Slide 18: Risk Neutral agent

$$\max_{w_0, w_1} p_1 (s_1 - w_1) + (1-p_1)(s_0 - w_0)$$

$$\text{s.t.} \quad \begin{aligned} p_1 w_1 + (1-p_1)w_0 - \psi &\geq 0 && \text{IR or PC} \\ p_1 w_1 + (1-p_1)w_0 - \psi &\geq p_0 w_1 + (1-p_0)w_0 && \text{IC} \\ \hookrightarrow (p_1 - p_0)(w_1 - w_0) - \psi &\geq 0 \end{aligned}$$

$$\mathcal{L} = p_1 (s_1 - w_1) + (1-p_1)(s_0 - w_0) + \lambda [p_1 w_1 + (1-p_1)w_0 - \psi] + \mu [(p_1 - p_0)(w_1 - w_0) - \psi]$$

F.O.C.:

$$\left. \begin{aligned} w_1: -p_1 + \lambda p_1 + \mu (p_1 - p_0) &= 0 \\ w_0: -(1-p_1) + \lambda (1-p_1) - \mu (p_1 - p_0) &= 0 \end{aligned} \right\} \Rightarrow \begin{aligned} p_1 &= \lambda p_1 + \mu (p_1 - p_0) \\ 1-p_1 &= \lambda (1-p_1) - \mu (p_1 - p_0) \end{aligned}$$

First result:
 $\lambda = 1$

$$\frac{1-p_1 = \lambda(1-p_1) - \mu(p_1-p_0)}{1} = \lambda \quad \text{add them up.}$$

$$\lambda \geq 0 \quad p_1 w_1 + (1-p_1)w_0 - \psi \geq 0, \quad \lambda [p_1 w_1 + (1-p_1)w_0 - \psi] = 0 \Rightarrow \lambda > 0 \Rightarrow p_1 w_1 + (1-p_1)w_0 = \psi$$

IR is binding.

$$\mu \geq 0 \quad (p_1 - p_0)(w_1 - w_0) - \psi \geq 0, \quad \mu [(p_1 - p_0)(w_1 - w_0) - \psi] = 0$$

$\lambda = 1 \Rightarrow$ From FOC for w_1 , $p_1 = p_1 + \mu(p_1 - p_0)$ since $p_1 > p_0 \Rightarrow \mu = 0 \Rightarrow$ IC is not necessarily binding.

∴ The solution is any (w_0, w_1) such that they satisfy $p_1 w_1 + (1-p_1) w_0 = \psi$ and $(p_1 - p_0)(w_1 - w_0) \geq \psi$.

Graphically: Note, no limited liability $(w_1, w_0) \in \mathbb{R}^2$,

The iso-profit line

$$\bar{\pi} = p_1(s_1 - w_1) + (1-p_1)(s_0 - w_0)$$

$$w_0 = \frac{p_1}{1-p_1}(s_1 - w_1) + s_0 - \frac{\bar{\pi}}{1-p_1}$$

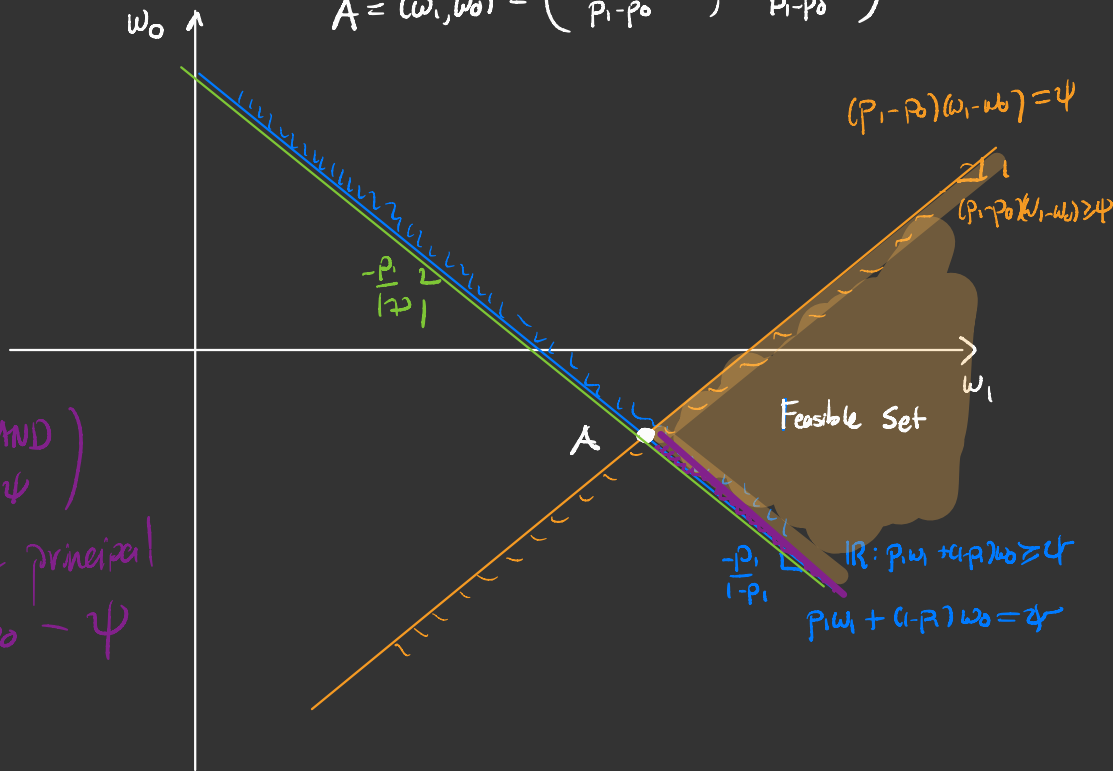
$$\text{slope of iso profit} = -\frac{p_1}{1-p_1}$$

The solution is the set

$$\{(w_1, w_0) \in \mathbb{R}^2 : p_1 w_1 + (1-p_1) w_0 = \psi \text{ AND } (p_1 - p_0)(w_1 - w_0) \geq \psi\}$$

The optimal profit for the principal is $\pi^* = p_1 s_1 + (1-p_1) s_0 - \psi$

$$A = (w_1, w_0) = \left(\frac{1-p_0}{p_1-p_0} \psi, -\frac{p_0}{p_1-p_0} \psi \right)$$

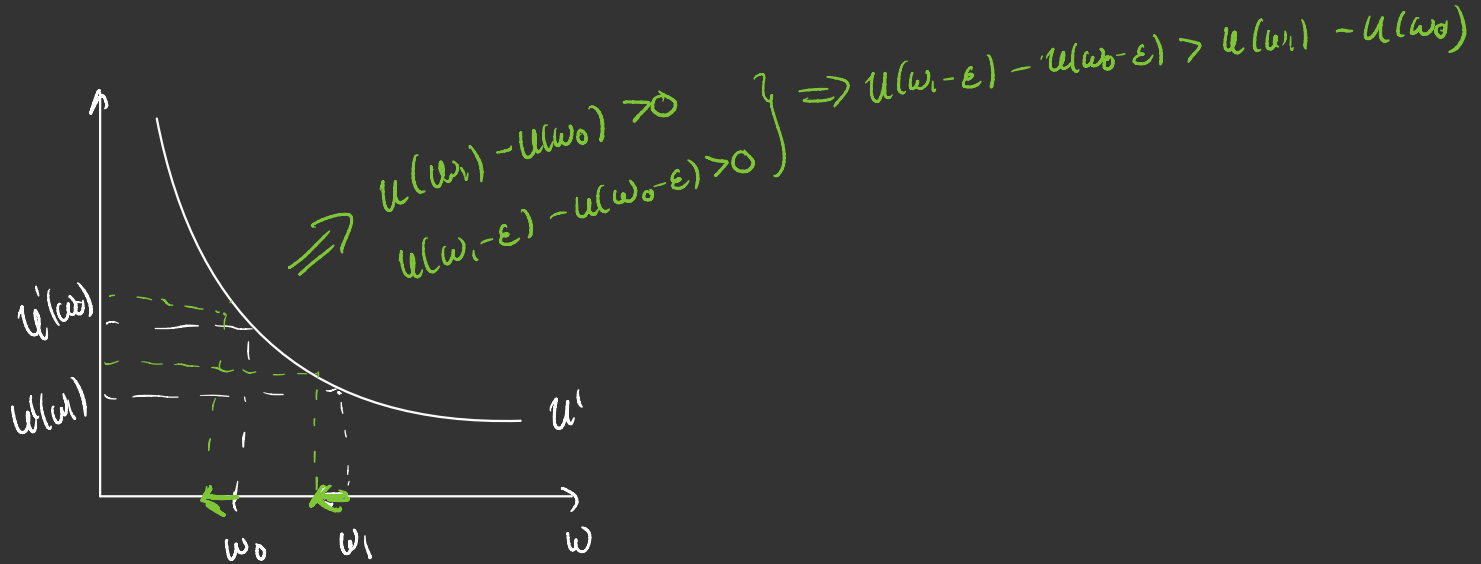


Notes on slide 21

IC constraint $(p_1 - p_0) [u(\omega_1) - u(\omega_0)] \geq \psi$

$\psi > 0 \Rightarrow u(\omega_1) > u(\omega_0)$ u is increasing ($u' > 0$)

$\Rightarrow \omega_1 > \omega_0$



Notes on 22

- Suppose $\lambda = 0 \Rightarrow \left. \begin{aligned} \frac{1}{u'(w_0)} &= \mu \\ \frac{1}{u'(w_1)} &= \mu \end{aligned} \right\} \Rightarrow \frac{1}{u'(w_0)} = \frac{1}{u'(w_1)} \Rightarrow u'(w_0) = u'(w_1) \left\{ \begin{array}{l} \text{Contradicts} \\ (3) \end{array} \right. \Rightarrow w_0 = w_1$

$\Rightarrow \lambda > 0$ / $\Rightarrow$ IC is binding

- Suppose $\mu = 0$, from FOC for w_0 :

$$\frac{1}{u'(w_0)} = -\lambda \frac{p_1 - p_0}{1 - p_0}, \quad \lambda \geq 0, \quad p_1 - p_0 > 0, \quad 1 - p_0 > 0$$

$$\underbrace{\frac{1}{u'(w_0)}}_{\leq 0} \leq 0 \quad \text{Contradiction} \Rightarrow \mu > 0$$

$$\Rightarrow \frac{1}{u'(w_0)} > 0 \Rightarrow \text{IR is binding.}$$

The solution is obtained by solving the non-linear system

$$(p_1 - p_0) [u(w_1) - u(w_0)] = \psi$$

$$p_1 u(w_1) + (1 - p_1) u(w_0) = \psi$$

$\Downarrow$
solve first for $u(w_0), u(w_1)$
then use $n = u^{-1}$ to
obtain the solution (w_0, w_1)

Notes on slide 25: for optimal schedule (w_0, w_1)

$$\psi < u(p_1 w_1 + (1-p_1) w_0) \Rightarrow h(\psi) < \underbrace{p_1 w_1 + (1-p_1) w_0}_{\text{cost of inducing effort at the second best}}$$

$\parallel$
cost of inducing effort under the first best

Notes on 28:

the fact that $w_1 > w_0$ implies that the problem becomes

$$\max_{w_0, w_1} p_1 (s_1 - w_1) + (1 - p_1) (s_0 - w_0)$$

s.t.

$$p_1 w_1 + (1 - p_1) w_0 - \psi \geq 0$$

$$(p_1 - p_0) (w_1 - w_0) - \psi \geq 0$$

$$w_0 \geq -l$$

Recall that without limited liability we found as one solution

$$w_0 = -\frac{p_0}{p_1 - p_0} \psi, w_1 = \frac{1 - p_0}{p_1 - p_0} \psi \quad (*)$$

was obtained by solving simultaneously IC and IR constraints as equalities.

Trick: solve the problem first without limited liability, by using solution (*), then check if it satisfies the constraint $w_0 \geq -l$ or not.

• Suppose that: $w_0 \geq -l \Leftrightarrow \frac{-p_0}{p_1 - p_0} \psi \geq -l$

$$\Leftrightarrow \frac{p_0}{p_1 - p_0} \psi \leq l$$

In this case IR binds, limited liability doesn't

and one solution is when IR & IC bind simultaneously:

$$\boxed{w_0 = \frac{-p_0}{p_1 - p_0} \psi \quad \text{and} \quad w_1 = \frac{1 - p_0}{p_1 - p_0} \psi}$$

• Suppose that: $w_0 < -l \Leftrightarrow \frac{-p_0}{p_1 - p_0} \psi < -l \Leftrightarrow \frac{p_0}{p_1 - p_0} \psi > l$

$\Rightarrow$ IC binds together with $w_0 = -l$

$$(p_1 - p_0)(w_1 + l) = \psi \Rightarrow \boxed{w_1 = \frac{\psi}{p_1 - p_0} - l}$$

Graphically

If $l \geq \frac{p_0 \psi}{(p_1 - p_0)}$

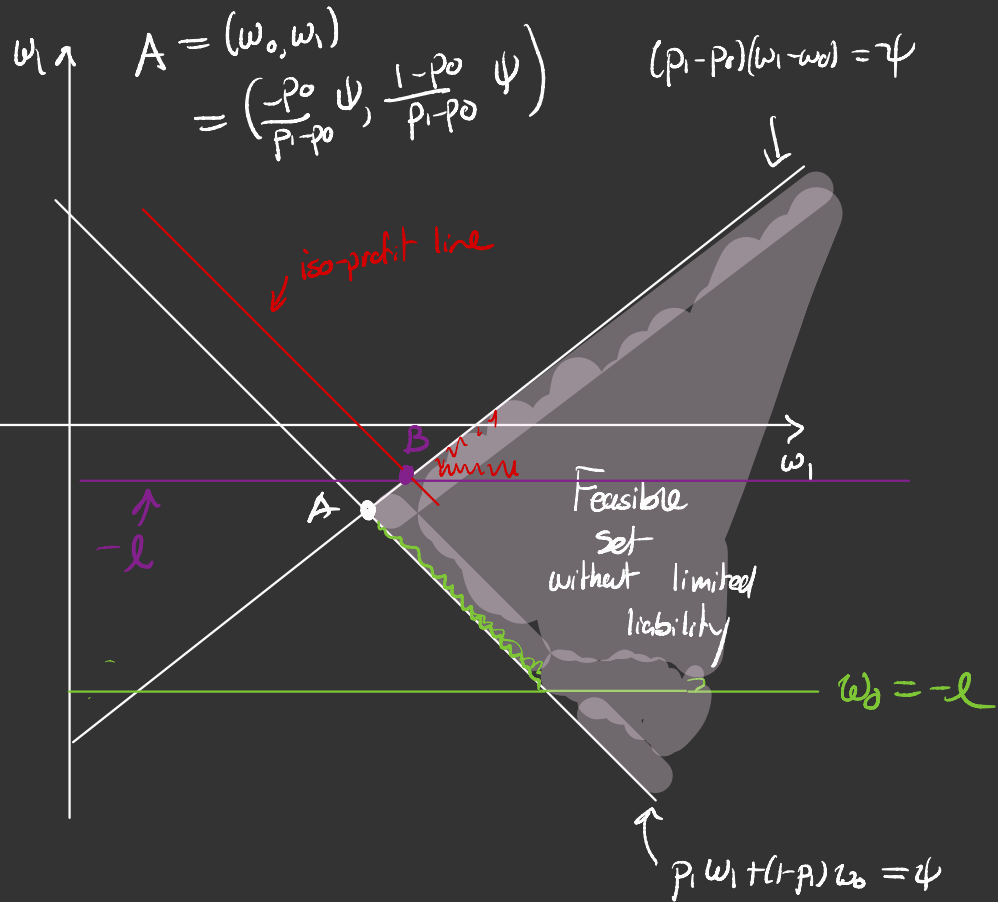
Sol. set is

$$\left\{ (w_1, w_0) : \begin{aligned} p_1 w_1 + (1-p_1) w_0 &= \psi \\ (p_1 - p_0)(w_1 - w_0) &\geq \psi \\ w_0 &\geq -l \end{aligned} \right\}$$

If $l < \frac{p_0 \psi}{(p_1 - p_0)}$

the solution is point B:

$$(w_0, w_1) = \left(-l, -l + \frac{\psi}{p_1 - p_0} \right)$$



Slide 33: Incomplete information, if the principal wants to induce effort.

$$\max_{(w_i)_{i=1}^n} \sum_{i=1}^n p_{i1} (s_i - w_i)$$

$$\text{s.t.} \quad \sum_{i=1}^n p_{i1} u(w_i) \geq \psi \quad \text{IR (Participation)}$$

$$\sum_{i=1}^n p_{i1} u(w_i) - \psi \geq \sum_{i=1}^n p_{i0} u(w_i) \quad \text{IC} \Leftrightarrow \sum_{i=1}^n [p_{i1} - p_{i0}] u(w_i) - \psi \geq 0$$

$$\mathcal{L} = \sum_{i=1}^n p_{i1} (s_i - w_i) + \mu \left[\sum_{i=1}^n p_{i1} u(w_i) - \psi \right] + \lambda \left[\sum_{i=1}^n (p_{i1} - p_{i0}) u(w_i) - \psi \right]$$

FOC:

$$\text{foreach } w_i: \quad -p_{i1} + \mu p_{i1} u'(w_i) + \lambda (p_{i1} - p_{i0}) u'(w_i) = 0$$

$$\Leftrightarrow u'(w_i) [\mu p_{i1} + \lambda (p_{i1} - p_{i0})] = p_{i1}$$

$$\Leftrightarrow u'(w_i) \left[\mu + \lambda \left(\frac{p_{i1} - p_{i0}}{p_{i1}} \right) \right] = 1$$

$$\Leftrightarrow \mu + \lambda \left(\frac{p_{i1} - p_{i0}}{p_{i1}} \right) = \frac{1}{u'(w_i)}$$

For each w_i :
$$\frac{1}{u'(w_i)} = \mu + \lambda \left(\frac{p_{i1} - p_{i0}}{p_{i1}} \right)$$

Multiplying by p_{i1} :
$$p_{i1} \left(\frac{1}{u'(w_i)} \right) = p_{i1} \mu + \lambda (p_{i1} - p_{i0})$$

both sides

adding up over i :
$$\sum_{i=1}^n p_{i1} \left(\frac{1}{u'(w_i)} \right) = \mu \sum_{i=1}^n p_{i1} + \lambda \left(\sum_{i=1}^n p_{i1} - \sum_{i=1}^n p_{i0} \right)$$

$$\Leftrightarrow \sum_{i=1}^n p_{i1} \left(\frac{1}{u'(w_i)} \right) = \mu + \lambda (1-1)^0$$

$$\Leftrightarrow \mu = \sum_{i=1}^n p_{i1} \left(\frac{1}{u'(w_i)} \right) = E \left[\frac{1}{u'(w_i)} \right] > 0$$

$\Rightarrow \mu > 0 \Rightarrow$ Participation
Constraint is
binding

expectation of the
inverse of the marginal
utility

IC also binds:

Suppose not, $\lambda = 0$, FOC's for each i :

$$\frac{1}{u'(w_i)} = \mu \Rightarrow u'(w_i) = u'(w_j) \text{ for } i, j \in \{1, \dots, n\}$$
$$\Rightarrow w_i = w_j \text{ for all } i, j. \text{ but this}$$

violates IC: $\sum_{i=1}^n (p_{i1} - p_{i0}) u(w_i) \geq \psi$

but if $w_i = w$ for all i $\Rightarrow \sum_{i=1}^n (p_{i1} - p_{i0}) u(w) = u(w) \sum_{i=1}^n (p_{i1} - p_{i0}) \overset{0}{=} 0 \geq \psi$!

contradiction, so λ cannot be 0.

$\therefore$ so $\lambda > 0$ and IC binds.